

LAPORAN PRAKTIKUM INTERNET OF THINGS 2



Oleh:

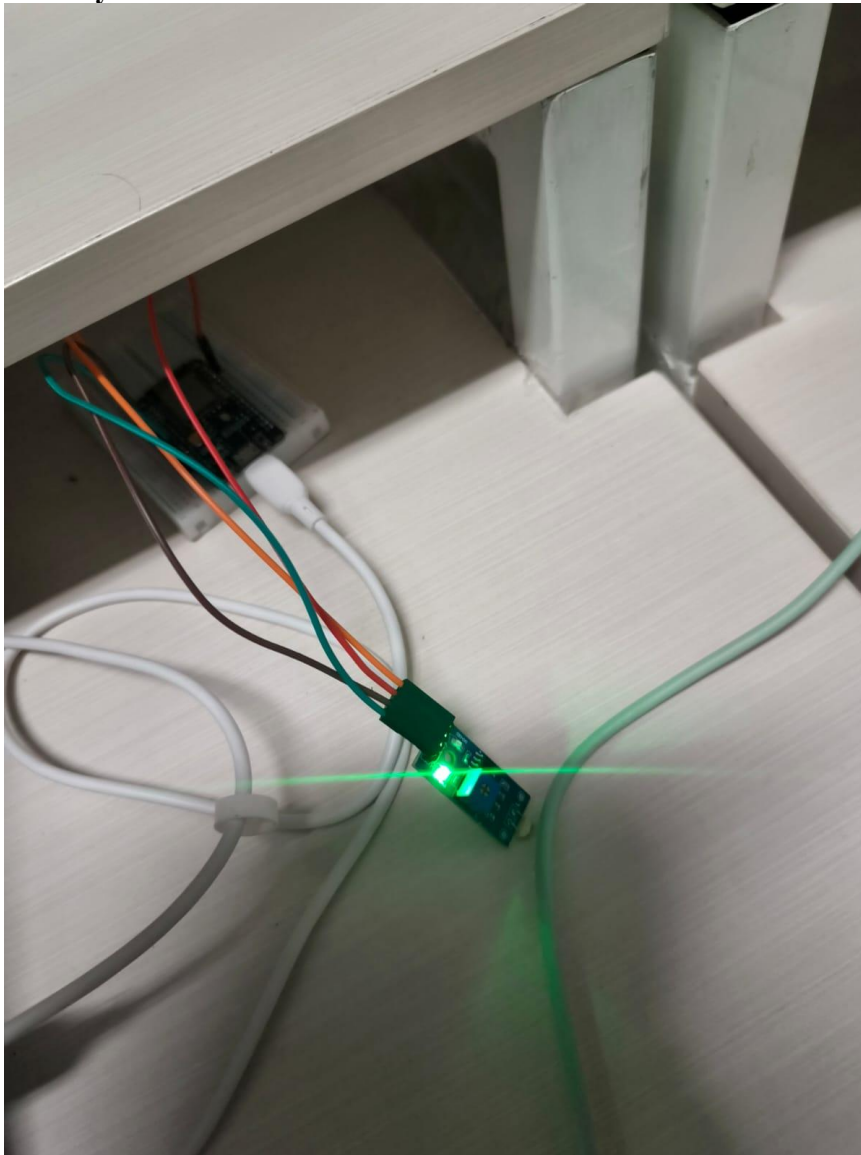
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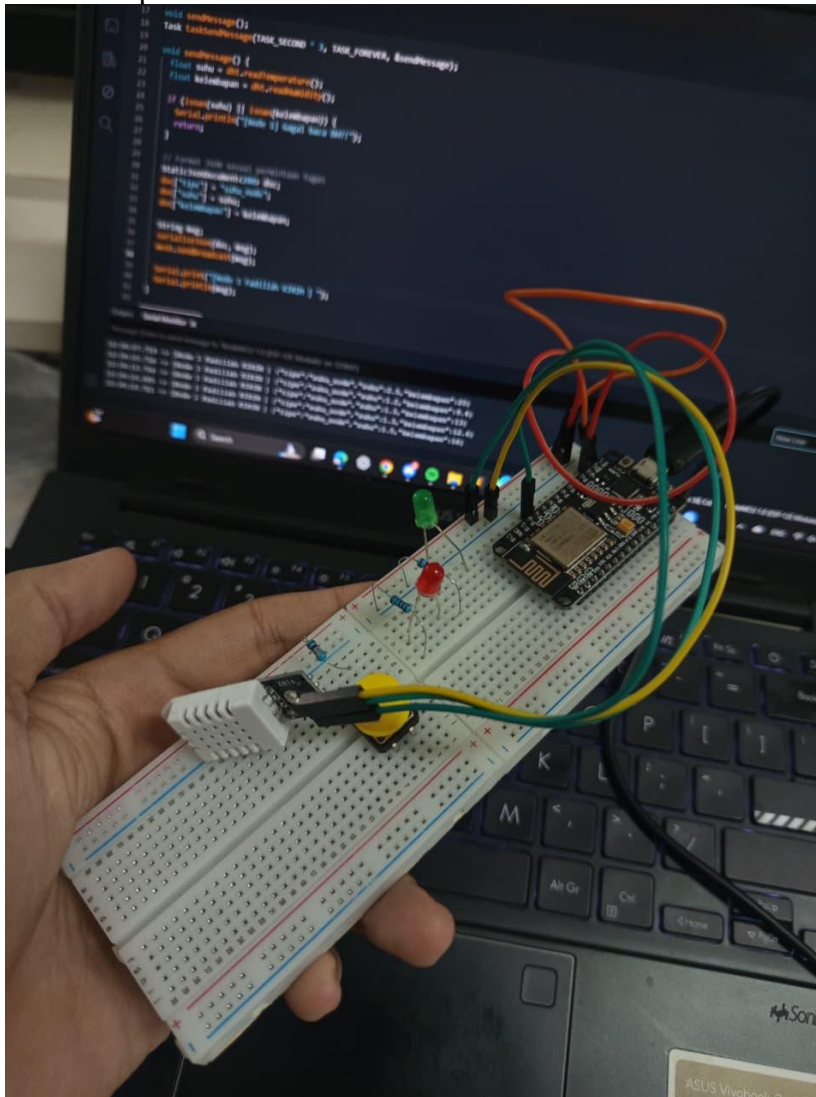
**TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI LAMPUNG
2026/2027**

5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh - Cahaya



```
sketch_oct5h | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...)
sketch_oct5hino
1 #include <painlessMesh.h>
2 #include <ArduinoJson.h>
3
4 #define MESH_PREFIX "LabIoTMesh"
5 #define MESH_PASSWORD "iotmeshpassword"
6 #define MESH_PORT 5555
7
8 #define LDRPIN A0 // Pin Analog
9
10 Scheduler userScheduler;
11 painlessMesh mesh;
12
13 void sendMessage();
14 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
15
16 void sendMessage() {
17   int nilaiADC = analogRead(LDRPIN);
18 }
Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM6')
[Node 2 KIRIM] ["type":"cahaya_node","adc":205]
[Node 2 KIRIM] ["type":"cahaya_node","adc":160]
[Node 2 KIRIM] ["type":"cahaya_node","adc":142]
[Node 2 KIRIM] ["type":"cahaya_node","adc":286]
[Node 2 KIRIM] ["type":"cahaya_node","adc":669]
[Node 2 KIRIM] ["type":"cahaya_node","adc":446]
[Node 2 KIRIM] ["type":"cahaya_node","adc":838]
Ln 12, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM6 2
ENG 12:35
UK 05/10/2026
```

- Kelembapan dan Suhu



```
sketch_oct5h | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...)
sketch_oct5h.ino
17 void sendMessage();
18 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
19
20 void sendMessage() {
21   float suhu = dht.readTemperature();
22   float kelembapan = dht.readHumidity();
23
24   if (isnan(suhu) || isnan(kelembapan)) {
25     Serial.println("[Node 1] Gagal baca DHT!");
26     return;
27   }
28
29   // Format JSON sesuai permintaan tugas
30   StaticJsonDocument<200> doc;
31   doc["type"] = "suhu_node";
32   doc["suhu"] = suhu;
33   doc["kelembapan"] = kelembapan;
34
35   String msg;
36   serializeJson(doc, msg);
37   mesh.sendBroadcast(msg);
38
39   Serial.print("[Node 1 Fadillah KIRIM ] ");
40   Serial.println(msg);
41 }
42
Output Serial Monitor X
Message [Enter to send message to 'NodeMCU 1.0 (ESP-12 Module)' on COM3]
12:34:07.719 -> [Node 1 Fadillah KIRIM ] {"type":"suhu_node","suhu":2.9,"kelembapan":20}
12:34:10.726 -> [Node 1 Fadillah KIRIM ] {"type":"suhu_node","suhu":1.2,"kelembapan":9.4}
12:34:13.704 -> [Node 1 Fadillah KIRIM ] {"type":"suhu_node","suhu":1.3,"kelembapan":13}
12:34:16.695 -> [Node 1 Fadillah KIRIM ] {"type":"suhu_node","suhu":1.3,"kelembapan":12.6}
12:34:19.721 -> [Node 1 Fadillah KIRIM ] {"type":"suhu_node","suhu":1.5,"kelembapan":16}
```

- Penerima

```
sketch_oct5ino | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12E)
sketch_oct5ino
47
48 Serial.print("[KIRIM RESH] ");
49 Serial.println(msg);
50
51
52 // callback otomatis saat menerima pesan dari node maupun di jaringan mesh
53 void receivedCallback(uint32_t from, String msg) {
54   // Parsing pesan type 2000 yang diterima
55   StaticJsonDocument<200> doc;
56   DeserializationError error = deserializeJson(doc, msg);
57
58   if (error) {
59     const char* sender = doc["node"];
60     uint32_t chipid = doc["chipid"];
61     float suhu = doc["suhu"];
62     float kelembapan = doc["kelembapan"];
63
64     Serial.println("-----");
65     Serial.print("[TERIMA DATA] Xs (Node ID: Xu | chip ID: Xu)\n", sender, from, chipid);
66     Serial.print("Suhu : %.2f °C\n", suhu);
67     Serial.print("Kelembapan : %.2f %\n", kelembapan);
68     Serial.println("-----");
69   } else {
70     Serial.print("[TERIMA DATA MENTAH DARI Xu]: Xu\n", from, msg.c_str());
71   }
72 }
73
74 // callback otomatis saat ada perangkat baru bergabung ke mesh
75
```

Output Serial Monitor X

Message (Enter to send message to NodeMCU 1.0 (ESP-12E Module) on COM15)

```
12:35:26.594 -> [TERIMA DATA] Xs (Node ID: Xu | chip ID: Xu)\n, sender, from, chipid);
12:35:29.174 -> [TERIMA DATA] Cahaya (ADC): 859 | STATUS: LED MATI (Kondisi Normal)
12:35:29.578 -> [TERIMA DATA] Suhu: 1.00 °C | STATUS: LED MATI (Kondisi Normal)
12:35:32.200 -> [TERIMA DATA] Cahaya (ADC): 987 | STATUS: LED MATI (Kondisi Normal)
12:35:32.573 -> [TERIMA DATA] Suhu: 2.50 °C | STATUS: LED MATI (Kondisi Normal)
```

Ln 22, Col 67 NodeMCU 1.0 (ESP-12E Module) on COM15

